CHARAN KUMAR SELVAM

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TECHNICAL SKILLS

Coding Languages : C, C++, C#, CUDA, **Python**.
Libraries : NumPy, SciPy, Pandas, Keras,**PyTorch, Pytorch Lightning, TensorFlow**, PyTorch3D, Kaolin, Diffusers, Matplotlib, Scikit-Learn, Open3D, OpenCV, FastAPI, Flask, PyTest, ONNXRuntime.
Databases : SQL, PostgreSQL, InfluxDBv2, NoSQL
Tools and Frameworks : Grafana, Git, Sagemaker, **LangChain, LangSmith, Dockers, MLFlow, RAG**, WandB.
Cloud Technologies : Amazon AWS, Microsoft Azure.
Relevant Skills and Courses : Machine Learning, Deep Learning, Computer Vision, Generative AI/LLM, Image Processing, 3D Processing, GPU programming, Prompt Engineering.
Certifications : **AWS Certified AI Practitioner**.

EXPERIENCE

Cavallo Technologies **May 2024 – Nov 2024**
Software Development Intern - C# | Microsoft Azure *Burnaby, Canada*

- Developing APIs using C# for banking solutions for the Client Coastal Community Bank, with continuous integration and deployment on Azure cloud along with CosmosDB and Azure Databricks.
- Developed **IaC** code to deploy Azure Resources using Bicep files. Also, debug and resolve CI/CD Azure pipeline issues.
- Leveraged Cypress scripts and Newman for API testing automation, and prior to integration with CI/CD pipelines.

Hypothetic **May 2023 – Jan 2024**
ML Research Intern (3D Computer Vision Focus) - Python | Computer Vision | ML | 3D AI | ***GPU*** *Vancouver, Canada*

- Worked on a research project aimed at developing and fine-tuning a **Deep Neural Network** architecture using **Pytorch** for predicting **Topological features** of 3D models.
- Developed a Deep Learning model leveraging diverse modalities to predict 3D model orientations with a 97% accuracy. Also, developed a Dockerized API integrating the model in **ONNX** format for CPU deployment.
- Enhanced a 3D alignment model by reducing intermediate **VGG-16** embedding size used for alignment correction, and reduced processing time by **40%** thereby improving efficiency and overall performance.

Extreme Networks **Aug 2019 – Aug 2022**
Associate Software Engineer - ***C | C++ | Python | Device Drivers | Linux kernel*** *Bengaluru, India*

- Integrated **6GHz** band(Wi-Fi 6E) support into modules within WingOS, for deployment on Access Points and Controllers using **C/C++ and Python**. Also supported maintenance and resolved Customer-reported Defects.
- Supported Synthetic NICs on **Cloud Controllers** when deployed on **Microsoft Azure** and **Hyper-V** instances.
- Designed and Prototyped an **pipeline** to harvest **network statistics** from APs and transmit them to **InfluxDB** using **Telegraf** for comprehensive real-time Network monitoring.

Renovus Vision Automation **Dec 2018 – Mar 2019**
ML Research Intern - **C++** | Python | **OpenGL** *Bengaluru, India*

- Integrating **Object Detection(YOLO)** algorithms for high-speed quality inspections, resulting in an **18%** reduction in processing time.
- Conducted research and survey on techniques for inspecting **3D Point Cloud** data and **defect detection**, overcoming occlusion and depth estimation challenges in 2D imagery.
- Developed a system to estimate dimensions using Point Clouds after **PointNet++ Part Segmentation** using **C++ and OpenGL**. Also gathered the Point Cloud Data, Labelled, and Augmented 3D data for training.

PROJECTS

StyleGAN Latent Vector Interpolation for Facial expression Editing | *Python, PyTorch, SVM* **Apr 2023**

- Identified an SVM Hyperplane in StyleGAN's Latent Vector feature Space separating Facial Expressions.
- Performed latent vector interpolation across hyperplane to modify facial expressions while preserving facial features.

Research Paper Summarizer | *LLM, LangChain, Python* **July 2023**

- Explored **RAG, LangChain, Pinecone, and GPT** to summarize research papers, and enable efficient semantic search to stay up-to-date on research.

Point Cloud Compass | *C++, OpenGL* **July 2019**

- Developed a software system capable of visualizing Point Clouds using OpenGL and C++, enabling **user-directed** navigation over its surface and accurately estimating dimensions of the 3D object.

Staging Diabetic Retinopathy(DR) using Retinal Images | *Python, OpenCV, K-NN* **Apr 2019**

- Designed a classifier model to assess DR severity from retinal images using Neural architecture and Image Processing achieving **94%** accuracy.

EDUCATION

M.Sc., Professional Computer Science - Visual Computing **Sep. 2022 – Apr. 2024**
Simon Fraser University *Burnaby, BC Canada*

B.E. in Information Science and Engineering **Jul. 2015 – Aug. 2019**
PES University *Bangalore, Karnataka, India*